



**SAVEETHA SCHOOL OF ENGINEERING**  
**SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES**  
**COMPUTER SCIENCE AND ENGINEERING PROGRAMME**  
**DSA0115 Object Oriented Programming with C++**  
**for System Level Programming**



**LEVEL 1 QUESTIONS**

1. Inverted right triangle pattern.

```
main.cpp
1 #include <iostream>
2 using namespace std;
3
4 int main() {
5     int n;
6     cout << "Enter number of rows: ";
7     cin >> n;
8
9     for (int i = n; i >= 1; i--) {
10         for (int j = 1; j <= i; j++) {
11             cout << "* ";
12         }
13         cout << endl;
14     }
15     return 0;
16 }
17
```

Output

```
Enter number of rows: 5
*****
****
***
**
*
```

==== Code Execution Successful ====

2. Check whether a number is a **Strong number**.

Strong

**STRONG**

Write a program to read in an integer and determine if it is a strong number or not.

```
1 #include <iostream>
2 using namespace std;
3 int main()
4 {
5     int n, temp, rem, sum = 0, fact;
6     cin >> n;
7     temp = n;
8     while(n > 0) {
9         rem = n % 10;
10        fact = 1;
11        for(int i = 1; i <= rem; i++)
12        {
13            fact = fact * i;
14        }
15        sum = sum + fact;
16        n = n / 10;
17    }
18    if(sum == temp)
19        cout << "Strong Number";
20    else
21        cout << "Not Strong Number";
22    return 0;
23 }
```

OUTPUT

```
Strong Number
```

145

### 3. Count the **frequency of each digit** in a number.

```
main.cpp  [Icons]  Share  Run  Output

3
4 int main() {
5     long long num;
6     int freq[10] = {0};
7
8     cout << "Enter a number: ";
9     cin >> num;
10
11     while (num > 0) {
12         int digit = num % 10;
13         freq[digit]++;
14         num /= 10;
15     }
16
17     cout << "Frequency of each digit:\n";
18     for (int i = 0; i <= 9; i++) {
19         if (freq[i] > 0)
20             cout << i << " : " << freq[i] << endl;
21     }
22
23     return 0;
24 }
```

Enter a number: 2345  
Frequency of each digit:  
2 : 1  
3 : 1  
4 : 1  
5 : 1  
  
== Code Execution Successful ==

### 4. Count the **number of divisors** of a number.

```
main.cpp  [Icons]  Share  Run  Output

1 #include <iostream>
2 using namespace std;
3
4 int main() {
5     int n, count = 0;
6
7     cout << "Enter a number: ";
8     cin >> n;
9
10    for (int i = 1; i <= n; i++) {
11        if (n % i == 0)
12            count++;
13    }
14
15    cout << "Number of divisors = " << count;
16
17    return 0;
18 }
```

Enter a number: 567  
Number of divisors = 10  
  
== Code Execution Successful ==

## 5. Calculate **compound interest**.

| main.cpp                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Output                                                                                                                                                          |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>1 #include &lt;iostream&gt; 2 #include &lt;cmath&gt; 3 using namespace std; 4 int main() { 5     double principal, rate, time, amount, ci; 6     cout &lt;&lt; "Enter Principal: "; 7     cin &gt;&gt; principal; 8     cout &lt;&lt; "Enter Rate (%): "; 9     cin &gt;&gt; rate; 10    cout &lt;&lt; "Enter Time (years): "; 11    cin &gt;&gt; time; 12    amount = principal * pow((1 + rate / 100), time); 13    ci = amount - principal; 14    cout &lt;&lt; "Compound Interest = " &lt;&lt; ci &lt;&lt; endl; 15    cout &lt;&lt; "Total Amount = " &lt;&lt; amount &lt;&lt; endl; 16    return 0; 17 }</pre> | <pre>Enter Principal: 100 Enter Rate (%): 1.9 Enter Time (years): 6 Compound Interest = 11.9554 Total Amount = 111.955  === Code Execution Successful ===</pre> |